

Q1.

(a) Calculate $I_{D_{n,sat}}$ and $I_{D_{p,sat}}$

NMOS Saturation Current:

$$I_{D_{n,sat}} = \frac{1}{2} \mu_n C'_{ox} (W/L)_n (V_{GS} - V_{Tn})^2$$

Given:

- $\mu_n C'_{ox} = 115 \mu A/V^2$
- $(W/L)_n = 2$
- $V_{GS} = V_{DD} = 2.5 V, V_{Tn} = 0.5 V$

$$I_{D_{n,sat}} = \frac{1}{2} \times 115 \times 2 \times (2.5 - 0.5)^2 = \frac{1}{2} \times 115 \times 2 \times (2.0)^2 = 115 \times 4 = \underline{460 \mu A}$$

PMOS Saturation Current:

$$I_{D_{p,sat}} = \frac{1}{2} \mu_p C'_{ox} (W/L)_p (V_{SG} - |V_{Tp}|)^2$$

Given:

- $\mu_p C'_{ox} = 30 \mu A/V^2$
- $(W/L)_p = 4$
- $V_{SG} = V_{DD} = 2.5 V, |V_{Tp}| = 0.6 V$

$$I_{D_{p,sat}} = \frac{1}{2} \times 30 \times 4 \times (2.5 - 0.6)^2 = 60 \times (1.9)^2 = 60 \times 3.61 = \underline{216.6 \mu A}$$

(1)

Calculate R_n and R_p (Effective Pull-down/Pull-up Resistance)

$$R_n = \frac{3}{4} \frac{V_{DD}}{I_{D_{n, sat}}} = \frac{3}{4} \times \frac{2.5}{460 \times 10^{-6}} = \frac{1.875}{460 \times 10^{-6}} \approx 4.07 \text{ k}\Omega$$

$$R_p = \frac{3}{4} \frac{V_{DD}}{I_{D_{p, sat}}} = \frac{3}{4} \times \frac{2.5}{216.6 \times 10^{-6}} = \frac{1.875}{216.6 \times 10^{-6}} \approx 8.65 \text{ k}\Omega \quad (1)$$

Calculate Delays t_{PHL} and t_{PLH}

Given load capacitance $C_L = 0.5 \text{ pF} = 0.5 \times 10^{-12} \text{ F}$:

$$t_{PHL} = 0.69 \cdot R_n \cdot C_L = 0.69 \times 4.07 \times 10^3 \times 0.5 \times 10^{-12} = 0.69 \times 2.035 \times 10^{-9} \approx 1.4 \text{ ns}$$

$$t_{PLH} = 0.69 \cdot R_p \cdot C_L = 0.69 \times 8.65 \times 10^3 \times 0.5 \times 10^{-12} = 0.69 \times 4.325 \times 10^{-9} \approx 2.98 \text{ ns} \quad (2)$$

(b) Maximum Output Frequency

$$T_{\text{period, min}} = t_{PHL} + t_{PLH} = 1.4 \text{ ns} + 2.98 \text{ ns} = 4.38 \text{ ns}$$

$$f_{\text{max}} = \frac{1}{T_{\text{period, min}}} = \frac{1}{4.38 \times 10^{-9}} \approx 228 \text{ MHz} \quad (2)$$

(c) : Dynamic Power Consumption

$$P_{\text{dyn}} = f_{\text{max}} \cdot C_L \cdot V_{DD}^2 = 228 \times 10^6 \cdot 0.5 \times 10^{-12} \cdot (2.5)^2 = 228 \times 0.5 \times 6.25 \times 10^{-6} = 712.5 \text{ }\mu\text{W} \quad (2)$$

(d) : Transistor Sizing for 25% Faster Delay

• Required delay: $t_{\text{new}} = 0.75 \times t_{\text{old}}$

• Required sizing: $(W/L)_{\text{new}} = \frac{4}{3} \times (W/L)_{\text{old}}$

$$\bullet \text{ NMOS: } 2 \rightarrow \frac{8}{3} \quad (1)$$

$$\bullet \text{ PMOS: } 4 \rightarrow \frac{16}{3} \quad (1)$$

• Dynamic power increases with increased maximum frequency, but power-delay product remains unchanged ($CLV_{DD}^2/2$).

(2+2)

Q2.

Solution (2).

$$(a) S = \frac{0.18 \mu m}{0.12 \mu m} = 1.5$$

Fixed voltage scaling.

$$A' = \frac{A}{(1.5)^2} = \frac{0.7 \text{ mm}^2}{2.25} = 0.311 \text{ mm}^2$$

$$P' = \frac{0.4 \text{ mW}}{1 \text{ MHz}} \times 100 \text{ MHz} = 40 \text{ mW}$$

$$\text{Power Density} = \left(\frac{P}{\text{Area}} \right) \times S^2 = \frac{40}{0.7} \times 2.25 = 128.57 \frac{\text{mW}}{\text{mm}^2}$$

— (3M)

(b) General scaling

$$V = \frac{1.8}{1.5} = 1.2$$

$$P' = \frac{40 \text{ mW}}{(1.02)^2} = 27.78 \text{ mW}$$

$$\left(\frac{P}{A} \right)' = \text{power density} = \frac{P}{A} \times \frac{S^2}{V^2} = \frac{40}{0.7} \times \frac{1.5 \times 1.5}{1.02 \times 1.02} = 89.28 \frac{\text{mW}}{\text{mm}^2}$$

— (3M)

$$(c) f' = S f = 100 \text{ MHz} \times 1.5 = 150 \text{ MHz}$$

Assuming dynamic power dominates,

$$P'_{150} = P'_{100} S = 27.78 \text{ mW} \times 1.5 = 41.67 \text{ mW}$$

— (3M)

$$\left(\frac{P_{150}}{A} \right)' = \left(\frac{P_{100}}{A} \right)' S = 89.29 \times 1.5 = 133.9 \text{ mW/mm}^2$$

$$(d) \left(\frac{P_{150}}{A} \right)' = \left(\frac{P_{100}}{V^2} \right) \left(\frac{S^2}{A} \right) S = \frac{P_{100}}{A} \quad [\text{condition that power density is same as original processor}]$$

$$\Rightarrow V^2 = S^3$$

$$\Rightarrow V = S^{3/2} = (1.5)^{3/2} = 1.837$$

$$\therefore V = \frac{1.8}{V'} = 1.837$$

$$\therefore V' \geq 1.0 \text{ V}$$

— (3M)

Q3.

$$V_{out} = V_{DD}$$

(i)

a) Calculate t_{PLH} , t_{PHL} & t_p

$$\mu_n C_{ox} = 280 \mu A/V^2$$

$$V_{tn} = 0.5V$$

$$t_{PLH} = 0.69 R_L C_L = 155 ns \quad (2M)$$

t_{PHL} using avg. current method

$$t_{PHL} = \frac{C_L \Delta V}{I_{avg}} = \frac{C_L V_{DD}/2}{I_{avg}} = \frac{3 \times 10^{-12} \times 1.25}{I_{avg}}$$

$$I_{avg} = \frac{1}{2} (I_{sat} + I_{lin})_{NMOS} \quad (2M)$$

$$I_{avg} = \frac{1}{2} \left(\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2 + \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) V_{ds} \right)$$

$$I_{avg} = \frac{1}{2} \left(\frac{1}{2} 280 \times 10^{-6} \times 3 (2.5 - 0.5)^2 + 280 \times 10^{-6} \times 3 (2.5 - 0.5) 1.25 \right)$$

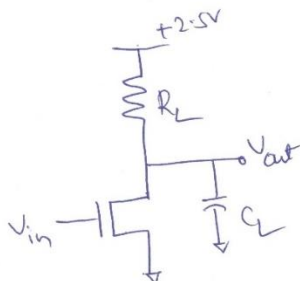
$$I_{avg} = \frac{1}{2} [1.68 + 1.44] \times 10^{-3} = 1.56 mA \quad (2M)$$

$$t_{PHL} = \frac{3 \times 10^{-12} \times 1.25}{1.56 mA} = 2.4 nsec \quad (3M)$$

$$t_p = \frac{t_{PLH} + t_{PHL}}{2} = 78.7 nsec$$

b)

$t_{PLH} \gg t_{PHL}$ because $R_L = 75 k\Omega$ is much higher than the effective linearized on-resistance of M_1 . (2M)



(1M)

Q4.

Solve (2) (a) $B=1$, given $H=1$

Design (a) $N=2$, $P=1+6=7$

$$G = (8/3)(1) = 8/3$$

$$\text{Delay } D = N(GH)^{1/N} + P$$

$$= 2(8/3)^{1/2} + 7$$

$$\boxed{D = 10.3 \tau} \quad \text{2 Marks}$$

Design (b) $N=2$, $P=3+2$,

$$G = 3/3 \times 5/3$$

$$\boxed{\text{Delay } D = 8.3 \tau} \quad \text{2 Marks}$$

Design (c) $N=2$, $P=2+3$,

$$G = (4/3)(7/3)$$

$$\boxed{\text{Delay } D = 8.5 \tau} \quad \text{2 Marks}$$

Design (d) $N=4$, $P=3+1+2+1=7$

$$G = 5/3 \times 1 \times 4/3 \times 1$$

$$\text{Delay} = N(GH)^{1/N} + P$$

$$\boxed{\text{Delay} = 11.8 \tau} \quad \text{2 Marks}$$

(b) Case (b) When $H=20$

Design	G	P	N	D ($H=20$)	
(a)	$8/3 \times 1$	$6+1$	2	21.6τ	1 Mark
(b)	$(5/3)(5/3)$	$3+2$	2	19.9τ	1 Mark
(c)	$(4/3)(7/3)$	$2+3$	2	20.8τ	1 Mark
(d)	$(5/3) \times 1 \times (4/3) \times 1$	$3+1+2+1$	4	17.3τ	1 Mark

for $H=20$, design (d) is best

$$\boxed{\text{Delay} = 17.3 \tau}$$

Q5.

Given that, $V_{T,load} = -3V$, $V_{T,driver} = 1V$, $V_{DD} = 5V$

Identify the worst case condition: 2 Marks

$A=1$, $D=0$, and either $B=1$ or $C=1$

Let, $A=1$, $D=0$, $B=1$ and $C=0$

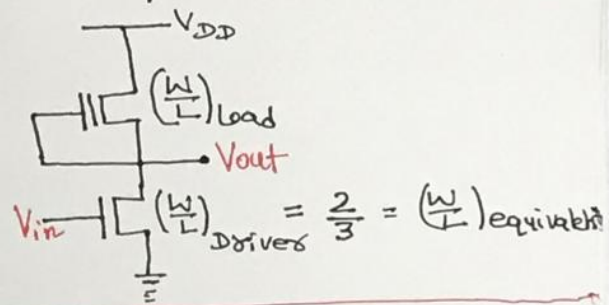
(This ensures that only one discharge path exists)

Simplify the circuit to an equivalent depletion load inverter: 2 Marks

$\therefore A=1$, $B=1$

$$\frac{1}{\left(\frac{W}{L}\right)_{\text{equivalent}}} = \frac{1}{\left(\frac{W}{L}\right)_A} + \frac{1}{\left(\frac{W}{L}\right)_B}$$

$$\Rightarrow \frac{1}{\left(\frac{W}{L}\right)_{\text{equivalent}}} = \frac{1}{\left(\frac{8}{4}\right)} + \frac{1}{\left(\frac{4}{4}\right)}$$



$$\Rightarrow \left(\frac{W}{L}\right)_{\text{equivalent}} = \frac{2}{3} = \left(\frac{W}{L}\right)_{\text{driver}} \text{ of the equivalent INVERTER}$$

Operating regions and governing equations:-

at $V_{OL} = 0.4V$, load transistor is in saturation and driver transistor is operating in linear region

$\Rightarrow I_{D,load} \text{ in saturation} = I_{D,driver} \text{ in linear}$ 2 Marks

$$\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_{load} (V_{DS,load} - V_{T,load})^2 = \mu_n C_{ox} \left(\frac{W}{L}\right)_{\text{equivalent}} [(V_{in} - V_{T,driver})V_{out} - \frac{1}{2} V_{out}^2]$$

$\therefore \mu_n C_{ox} = K'$, $V_{out} = V_O = V_{OL}$ 2 Marks

$$\Rightarrow \frac{K'}{2} \left(\frac{W}{L}\right)_{load} [0 - (-3)]^2 = K' \left(\frac{2}{3}\right) [(5-1)(0.4) - \frac{1}{2} (0.4)^2]$$

$$\Rightarrow \frac{K'}{2} \left(\frac{W}{L}\right)_{load} (9) = K' \left(\frac{2}{3}\right) [1.6 - 0.08]$$

$$\Rightarrow \left(\frac{W}{L}\right)_{load} = 0.225 \text{ for } V_{OL} \leq 0.4V$$

4 Marks

Q6.

